AI Driven Autonomous Steering

Autonomous Steering Wheel Using IoT & ML

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ABSTRACT

Overview

This project automates hardware steering actuation by extracting visual lane geometry using CNNs. It converts raw camera feeds into mechanically safe motor predictions instantaneously.

Methodology

- **CNN Architecture:** CNN (5 Conv + 4 Dense layers) to extrapolate spatial coordinates.
- **Image Preprocessing:** Localized skybox cropping and strict frame normalization (66x200 RGB).
- **Model Training:** Adam optimization (1e-4) combined with L2 constraints to minimize MSE.
- **Integration:** PySerial binding to Arduino stepper motors using 5-frame rolling queues.

Applications

Low-cost Driver Assistance Systems (ADAS), automated guided vehicles (AGVs), open-source research platforms, and automotive education environments.

PROBLEM STATEMENT & ANALYTICAL STUDY

Identified Gaps in Autonomous AI:

- Most retail self-driving models lack a functional, real-world hardware execution bridge.
- Existing physical prototypes struggle with extreme steering jitter and sudden motor burnout.
- High costs of proprietary LiDAR/RADAR make robotics prohibitive for small developers.

Proposed Solution Pipeline:

Video Intake → Normalization → CNN Inference → Jitter Filter → Stepper Actuation

Key Differentiator:

Fully offline-capable execution. It successfully guarantees reliable hardware steering exclusively through a basic video and a standard Arduino module, completely eliminating external API latency.

SYSTEM ARCHITECTURE & PROJECT SCOPE

Frontend / Optics Layer

- **Video Ingestion**
- Real-Time Video Streaming
- PyQt5 Interface
- Live Steering Widget

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Data Gateway

- **Target Parsing**
- Skybox Crop (-150px bounds)
- Resolution Transform (66x200)
- RGB Matrix Normalization

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AI / ML Processing

- **TensorFlow & CNN**
- 5 Conv + 4 Dense Layers

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Output Services

- **Motor Actuation**
- PySerial COM Port Binding
- 5 Frame Smoothing Filter
- Arduino Stepper Motor

Process Flow: OpenCV Intake → Matrix Normalization → CNN Inference → Rolling Average → Arduino Command

MACHINE LEARNING MODEL PERFORMANCE

Bar height = physical oscillation variance. The Combined approach guarantees stability.

1e-4
ADAM LEARNING RATE

66x200
INPUT FRAME GEOMETRY

< 3°
JITTER AVOIDANCE

5 Avg
ROLLING QUEUE BUFFER

ACTUATION BUILT-IN FEATURES

- Actuation Logic
- Jitter Proof
- Serial Linking
- Stepper Motor
- Spike Denial
- GUI Dashboard

Structured Integration Flow

- **Rolling Queue Mitigation** — Buffers 5 frames, calculating a localized average to heavily dampen noisy predictions.
- **Jitter Nullifier** — Explicit firmware logic ignores minor geometric variances (<3° difference) preventing micro-movements.
- **Spike Refusal** — The communication buffer aggressively denies single-frame torque spikes exceeding safe bounds.

Tech Stack Highlights:
- **Frontend:** PyQt5,Computer Vision (OpenCV)
- **Backend:** Python,PySerial Integration
- **AI / ML:** TensorFlow,Keras, CNN Regressors
- **Hardware:** ArduinoUNO, Stepper Motor

CONCLUSION

This project fully realizes an integrated, low-cost autonomous steering pipeline connecting raw camera optics to physical hardware. Key achievements include:

- **Scalable Cost Efficiency:** Operates entirely on consumer-grade webcams and IoT actuators, avoiding massive LiDAR infrastructure.
- **Offline Operation:** Checkpoints function locally with no required external network connectivity.
- **Mechanical Assurance:** Deep filter algorithms dramatically extend motor lifespan by suppressing inherent AI jitter.
- **RapidExecution:** Optimized CV scaling and Adam optimizers permit immediate safety response times.

FUTURE SCOPE & REFERENCES

Future Enhancements:

- Inclusion of dynamic object detection bounding pipelines alongside active steering predictors.
- Integration of Live PID loop control for automated braking and adaptive velocity.
- Deployment into modular plug-and-play kits enabling rapid academic robotic scaling.

References:

- Bojarski, M.etal. (2016). "End to End Learning for Self-Driving Cars". Nvidia Corporation. arXiv:1604.07316.
- Golluri Ramcharan and Dhunnapothula Raghu, "Arduino based stepper motor speed regulation for robot", Int. Journal / Conference Paper.